CORPORATE RISK MATRIX

Term	Rare	Unlikely	Possible	Likely	Very Likely	Almost Certain
Frequency	≥ 10^{-6} to < 10^{-5}	≥ 10^{-5} to < 10^{-4}	≥ 10^{-4} to < 10^{-3}	≥ 10^{-3} to < 10^{-2}	≥ 10^{-2} to < 10^{-1}	≥ 10^{-1} to ≤ 1
Probability	≥ 0.0001% to < 0.001%	≥ 0.001% to < 0.01%	≥ 0.01% to < 0.1%	≥ 0.1% to < 1%	≥ 1% to < 10%	≥ 10% to ≤ 100%
Likelihood	A similar event has not yet occurred somewhere in our industry	A similar event has occurred somewhere in our industry but not in ENTITIES	A similar event has occurred at least Once in ENTITIES	Event has occurred at least once in ENTITIES or is likely to occur within the lifetime of 10 similar COMPANIES	Event has occurred more than several times in ENTITIES or has occurred once or twice in the lifetime of the COMPANY	The event likely to occur several times in the lifetime of the COMPANY

Severity	HSE Impact		Business Impact		#	1	2	3	4	5	6
	Health and Safety	Environmental	Financial	Non-Financial							
			Financial Impact on Assets, Projects and Production	Reputation, Media, Key Stakeholder Reaction and Regulatory Enforcement							
Disastrous	• Potential for multiple fatalities Onsite. • > 10 lost-time injuries requiring hospital treatment for more than 24 hours. • Single or more than 1 Public Fatality • Multiple exposures to a hazardous substance or condition causing an emergency health effects that may threaten the lives of more than one worker.	• Environmental impact at the global level, e.g., the unintended release of hydrocarbons with globally widespread environmental damage. • Future unsatisfactory impact for a period > 5 years. • Discharge of contaminants > 100,000L in the insensitive environment and/or > 10,000L in the sensitive environment, e.g., coral reefs, nature reserves, rivers, canals, waterways, and close to groundwater wells.	• Catastrophic damage. • Revamping necessary to resume the operations/business (> 50% of Annual Production/Business/ Profit Plan).	• International concern. • Extensive negative attention in international media. • Long-term national outrage. • Substantial long-term damage to relationships with key stakeholders. • Action by a government resulting in a complete and permanent loss of license to operate. • Potentially severe impact on access to new areas.	A	1A	2A	3A	4A	5A	6A
Catastrophic	• Potential single fatality Onsite. • 4 to 10 lost-time injuries requiring hospital treatment for more than 24 hours. • Serious injury to the public. • Single exposure to a hazardous substance or condition causing emergency health effects that may threaten the life of one worker.	• Environmental impact at the national level, e.g., the unintended release of hydrocarbons with nationally widespread environmental damage. • Future unsatisfactory impact for more than one and up to five years. • Discharge of contaminants >10,000: ≤100,000 liters in the insensitive environment and/or >1,000: ≤10,000 liters in the sensitive environment, e.g., coral reefs, nature reserves, rivers, canals, waterways, and close to groundwater wells.	• Extensive damage. • Major change to resume operations/business (> 35%: ≤ 50% of Annual Production/Business/Profit Plan).	• International public attention. • National & global media coverage. • Medium-term national outrage. • Moderate long-term Damage to relationships with key stakeholders. • Action by a government resulting in partial and/or temporary loss of license to operate. • Substantial regulatory enforcement action by regulators.	B	1B	2B	3B	4B	5B	6B
Major	• 2 or 3 lost-time injuries cases. • A total irreversible disability case. • Multiple partial irreversible disabilities. • Single or multiple exposures to a hazardous substance or condition causing a state of ill health that extend for long periods.	• Environmental impact at the regional level, e.g., the unintended release of hydrocarbons with extensive environmental damage across several governances. • Future unsatisfactory impact for around one year. • Discharge of contaminants >1,000: ≤10,000 liters in the insensitive environment and/or >100: ≤1,000 liters in the sensitive environment, e.g., coral reefs, nature reserves, rivers, canals, and waterways, and close to groundwater wells.	• Major damage. • Long time change to resume operations/business (> 20%: ≤ 35% of Annual Production/Business/Profit Plan).	• National public concern. • Extensive negative attention in national media. • Long-term local outrage. • Partial Damage to relationships with key stakeholders. • Moderate regulatory enforcement action by regulators. • Potentially restrictive measures and/or impact on the grant of licenses.	C	1C	2C	3C	4C	5C	6C
Serious	• A Lost Time Injury case. • A partial irreversible disability case. • An injury that led to a reversible disability (total or partial). • Single or multiple exposures to a hazardous substance or condition causing a state of ill health requiring treatment > 24 hrs.	• Environmental impact at the local level, e.g., the unintended release of hydrocarbons with localized environmental damage within one governance. • Future unsatisfactory impact for months "less than one year). • Discharge of contaminants >100: ≤1,000 liters in the insensitive environment and/or >10: ≤100 liters in the sensitive environment, e.g., coral reefs, nature reserves, rivers, canals, waterways, and close to groundwater wells.	• Local damage. • The unit has been repaired/replaced to resume operations/business (> 1%: ≤ 20% of Annual Production /Business /Profit Plan).	• Regional public concern. • Negative attention from national media. • Limited local outrage. • Limited damage to relationships with local stakeholders. • Limited regulatory enforcement action by regulators. • Possibly negative stance of local government.	D	1D	2D	3D	4D	5D	6D
Minor	• Medical Treatment Case. • Restricted Work Day Case. • Single or multiple exposures to a hazardous substance or condition causing minor health effects requiring treatment < 24 hrs.	• Environmental impact at the immediate level, e.g., the unintended release of hydrocarbons with immediate area damage to the environment. • Future unsatisfactory impact for less than six months. • Discharge of contaminants >10: ≤100 liters in the insensitive environment and/or ≤10 liters in the sensitive environment, e.g., coral reefs, nature reserves, rivers, canals, waterways, and close to groundwater wells.	• Minor damage. • Possible short disruption of operations/business (> 0.1%: ≤ 1% of Annual Production / Business / Profit Plan).	• Some local public concern/media coverage. • Limited damage to relationships with local stakeholders. • Government reaction to non-compliance with legal and regulatory requirements that do not result in consequences beyond simply restoring regulatory compliance.	E	1E	2E	3E	4E	5E	6E
Notable	• First Aid Case. • Single or multiple exposures to a hazardous substance or condition causing a slight non-prolonged health effect.	• The limited impact on the environment at the level of the polluted area, e.g., the limited unintended release of hydrocarbons. • Future unsatisfactory impact for days or weeks. • Discharge contaminants with no impact on the insensitive environment, e.g., coral reefs, nature reserves, rivers, canals, and waterways, and close to groundwater wells, and/or ≤ 10 liters in an insensitive environment.	• Slight damage. • Disruption to operations/business (≤ 0.1% of Annual Production / Business / Profit Plan).	• No public concern/media coverage. • Negligible damage to relationships with local stakeholders. • Non-compliance with regulatory requirements that do not generate any government reaction.	F	1F	2F	3F	4F	5F	6F

Risk Level	Minimum Required Action	Management Signoff Authority
Level 1 - High (Intolerable)	Long-term operation at high risk is not allowed. Immediately identify risk mitigation according to the hierarchy of controls and implement risk mitigation to reduce the risk to ALARP within 30 days. In case of an extension, approval from the relevant holding company shall be obtained. Include in Risk Register for tracking. Consider advanced Risk Analysis methodologies for further investigation. Reporting the risk to the relevant top management within 24hrs. Notify EGPC or the relevant Holding Companies about the risk within 7 Days.	Signoff by Chairman / Managing Director(s) / VP's/CEO (Consultation with appropriate SME required)
Level 2 - Medium/High (Tolerable if ALARP)	Conduct an ALARP study to determine if the cost of implementing additional risk mitigation is appropriate. Include in Risk Register for tracking. Consider advanced Risk Analysis methodologies for further investigation. Reporting the risk to the relevant top management within 7 Days.	Signoff by Senior management, i.e., Operations GM, Projects GM, Drilling GM, Financial GM, HSE GM …etc.
Level 3 - Medium (Tolerable if ALARP)	Conduct an ALARP study to determine if the cost of implementing additional risk mitigation is appropriate. Monitor and improve the effectiveness of current controls. Include in Risk Register for tracking. Reporting the risk to the relevant top management within 30 Days.	Signoff by Function GM, i.e., District GM, Refinery Unit GM …etc.
Low (Broadly acceptable)	Low Priority, monitor and improve the effectiveness of current controls. Include in Risk Register for tracking.	Signoff by Area Authority, i.e., OIM, Production Manager, Maintenance Manager …etc.

Misleading Multiplications Matrix

		Likelihood					
Severity		1	2	3	4	5	6
A	6	6	12	18	24	30	36
B	5	5	10	15	20	25	30
C	4	4	8	12	16	20	24
D	3	3	6	9	12	15	18
E	2	2	4	6	8	10	12
F	1	1	2	3	4	5	6